



# Advanced Time Series Analysis and Forecasting

## Chapter 0: Refresher and inference for dependent data



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## Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **ALFRED**: Archival Federal Reserve Economic Data
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARDL**: AutoRegressive Distributed Lag (model)
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **ASDS**: Statistică aplicată și data science (the Master's programme Applied Statistics and Data Science)
- ▣ **ASE**: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ **ATS**: Advanced Time Series Analysis and Forecasting
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BVAR**: Bayesian Vector AutoRegression
- ▣ **CBB**: Circular Block Bootstrap
- ▣ **CLT**: Central Limit Theorem
- ▣ **CPI**: Consumer Price Index
- ▣ **DGP**: Data-Generating Process
- ▣ **ECB**: European Central Bank
- ▣ **ECTS**: European Credit Transfer and Accumulation System
- ▣ **EOD**: End Of Day (daily closing data)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **ES**: Expected Shortfall
- ▣ **EU**: European Union
- ▣ **EWC**: Equal-Weighted Cosine (estimator)
- ▣ **FDR**: False Discovery Rate
- ▣ **FM-OLS**: Fully Modified Ordinary Least Squares

## Acronyms used in this material (2/2)

- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP**: Gross Domestic Product
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- ▣ **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- ▣ **LLN**: Law of Large Numbers
- ▣ **LLSW**: Lazarus, Lewis, Stock and Watson (2018)
- ▣ **MA**: Moving Average
- ▣ **MBB**: Moving-Block Bootstrap
- ▣ **MDS**: Martingale Difference Sequence
- ▣ **MFM**: Modelling Financial Markets
- ▣ **MSE**: Mean Squared Error
- ▣ **NW**: Newey–West (standard errors)
- ▣ **NW-A**: Newey–West with the Andrews (1991) bandwidth
- ▣ **OLS**: Ordinary Least Squares
- ▣ **QR**: Quick Response (code)
- ▣ **QS**: Quadratic Spectral (kernel)
- ▣ **SARIMA**: Seasonal ARIMA (model)
- ▣ **SPA**: Superior Predictive Ability (test)
- ▣ **TSA**: Time Series Analysis
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

## The course at a glance

### □ Programme

- ▶ Master's programme Applied Statistics and Data Science (ASDS), year 1, semester 2
- ▶ Faculty of Cybernetics, Statistics and Economic Informatics, ASE Bucharest
- ▶ 7 ECTS credits; 2 hours of lecture and 2 hours of seminar a week; academic year 2026/2027

### □ Lecturer

- ▶ Prof. Daniel Traian Pele, [danpele@ase.ro](mailto:danpele@ase.ro)

### □ Course website

- ▶ [danpele.github.io/Advanced-Time-Series](https://danpele.github.io/Advanced-Time-Series)
- ▶ slides, seminars, notebooks, Quantlets and the graded quizzes



The Bucharest University of Economic Studies (ASE)

Photo: Joe Mabel (2014); CC BY 3.0; Wikimedia Commons

## Position in the curriculum

- **Before:** [Time Series Analysis](#) (TSA, bachelor), Chapters 0–10 of TSA
  - ▶ ARMA, ARIMA and unit roots, SARIMA, GARCH, VAR and Granger causality, cointegration and VECM, long memory, machine learning, state space models
  - ▶ assumed known: this course starts from them and does not repeat them
- **This course:** ATS, master, year 1, semester 2
  - ▶ methodology for forecasting, identification and inference with dependent data
  - ▶ macroeconomic (Romanian and EU), energy, climate and financial series
- **After:** [Modelling Financial Markets](#) (MFM, master, year 2, semester 1)
  - ▶ the same methods applied to market risk, portfolios, microstructure and digital assets

## Evaluation

- ▣ **Team project and individual oral defence: 70%**
  - ▶ proposal and pre-registration (team): 5%
  - ▶ replication and extension: repository, report, AI\_USE.md and AI\_ERRORS.md (team): 15%
  - ▶ individual oral defence: 50%
- ▣ **Chapter quizzes: 20%**
  - ▶ online, on the course website, with the ASE Google account
- ▣ **Attendance: 10%**
  - ▶ recorded with the attendance form or the QR code
- ▣ There is no written exam

## Textbooks

### ▣ Main references

- ▶ Hamilton (1994): the theory of linear time series, unit roots and VAR
- ▶ Kilian and Lütkepohl (2017): structural VAR analysis and identification
- ▶ Durbin and Koopman (2012): state space methods
- ▶ Petropoulos et al. (2022): forecasting theory and practice, open access

### ▣ Bridge from the bachelor course

- ▶ Huang and Petukhina (2022); Hyndman and Athanasopoulos (2021)

### ▣ Specialised

- ▶ Lütkepohl (2005); Shumway and Stoffer (2017); Baltagi (2021); Angelopoulos and Bates (2023); for this chapter also Lahiri (2003)
- ▶ each chapter adds its landmark papers as case studies

## Course map (1/2)

| Ch. | Topic                                                                        |
|-----|------------------------------------------------------------------------------|
| 0   | Refresher and inference for dependent data                                   |
| 1   | Forecast evaluation, scoring rules and combination                           |
| 2   | Structural breaks and nonlinear models                                       |
| 3   | Structural VAR and local projections                                         |
| 4   | Cointegration revisited: VECM, ARDL and panel data                           |
| 5   | Bayesian VAR, factor models and nowcasting                                   |
| 6   | State space models and Bayesian filtering                                    |
| 7   | Regime-switching models                                                      |
| 8   | Advanced volatility modelling: realised measures, HAR and multivariate GARCH |

## Course map (2/2)

| Ch. | Topic                                                          |
|-----|----------------------------------------------------------------|
| 9   | VaR, ES and backtesting: elicatability, scoring and model risk |
| 10  | Long memory and rough volatility                               |
| 11  | Spectral and wavelet analysis                                  |
| 12  | Machine learning and deep learning for time series             |
| 13  | Foundation models and conformal prediction                     |
| 14  | Causal inference for time series                               |
| 15  | Review and project defence                                     |
| 16  | Explosive roots and bubbles (self-study)                       |

## Lectures and seminars

- ▣ **Lectures**

- ▶ one chapter a week, built around landmark papers replicated on our data
- ▶ each lecture ends with a section on AI for scientific discovery

- ▣ **Seminars:** each seminar comes *before* its lecture and opens with the notions it needs

- ▶ Part A: derivations; Part B: estimation and inference on data, each exercise closing with an interpretation question; Part C: open problems that can seed the project

**Solved** exercises with full solutions and [Proposed] exercises discussed in class

- ▶ every seminar includes the critique of an answer given by an AI tool

- ▣ Seminars are practice and are not graded; the solutions of the [Proposed] exercises are discussed in class

## The team project

- **Teams of up to 3 students**
  - ▶ replicate a landmark paper of the course, or another paper approved by the lecturer
  - ▶ extend it with the methods of the course, preferably on Romanian or EU data
- **Stages**
  - ▶ Stage 1: research question, data and an evaluation design fixed before any estimation (pre-registration)
  - ▶ Stage 2: replication, extension and a GitHub repository that reproduces every number (Colab)
  - ▶ Stage 3: presentation and individual oral defence
- Inference that is valid for dependent data (this chapter) is part of every project
- The project can be written and defended in Romanian or in English

## AI policy

- ▣ **Allowed and declared**

- ▶ AI\_USE.md: tool, purpose, prompts, what was kept and what was changed
- ▶ AI\_ERRORS.md: at least three errors of AI tools caught by the team, and how each was detected
- ▶ undeclared use counts as plagiarism

- ▣ **Checked**

- ▶ every reference against its DOI, every number against the data and the code

- ▣ **Oral defence without AI**

- ▶ each member explains the methods, the code and the results

## Materials and tools

### ▣ For every chapter

- ▶ lecture and seminar slides in English and Romanian
- ▶ lecture and seminar notebooks in English, runnable in Google Colab
- ▶ a Quantlet with the code of every chart, and a quiz

### ▣ Data

- ▶ daily market data from EODHD (EOD Historical Data), saved in the course repository
- ▶ public macroeconomic data: INS, BNR, Eurostat, the ECB and FRED

### ▣ Software

- ▶ Python: statsmodels, arch, linearmodels, scikit-learn, PyTorch

## Learning outcomes

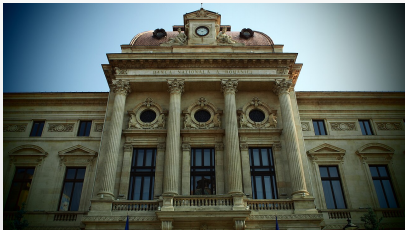
- ▣ State the limit theorems behind every standard error used with time series: ergodic LLN, martingale-difference and mixing CLT
- ▣ Derive the long-run variance and its consequences for  $t$ -tests and confidence intervals
- ▣ Choose and defend a HAC estimator: kernel, bandwidth, critical values (standard or fixed- $b$ )
- ▣ Use block bootstraps (moving, circular, stationary) and know when the wild bootstrap is not enough
- ▣ Measure size distortions by Monte Carlo and control data snooping
- ▣ Build a replication package that reproduces every number of a project

## Data used in this chapter

| Series             | Definition                                                  | Period                          | Source   |
|--------------------|-------------------------------------------------------------|---------------------------------|----------|
| Romanian inflation | HICP, annual rate of change, %, monthly                     | January 2006 – August 2026      | Eurostat |
| Romanian real GDP  | chain-linked volumes, seasonally adjusted; growth $q/q$ , % | 2000Q1 – 2026Q2                 | Eurostat |
| EUR/RON            | BNR reference rate; daily log change, %                     | 5 July 2005 – 18 September 2026 | BNR      |
| S&P 500, BET       | daily log returns and squared returns, %                    | 2000 – 18 September 2026        | EODHD    |
| US macro           | real GDP; 10-year and 3-month Treasury yields               | 1962 – 2026                     | FRED     |

- Daily series on their own trading calendars; Eurostat and FRED series read online at the build date
- HICP: harmonised index of consumer prices; the BNR target is defined on the national CPI, close to the HICP

## Where the data come from



The National Bank of Romania (BNR), Bucharest: reference rates and the inflation target

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- A central bank publishes the series we test: the inflation target of the BNR is  $2.5\% \pm 1$  pp since 2013 (the AI mini-case at the end)



The Eccles Building, Washington: the Federal Reserve Board;  
FRED is run by the St. Louis Fed

Photo: Federal Reserve (2011); public domain; Wikimedia Commons

## Notation (1/2)

- $\{x_t\}$ ,  $t = 1, \dots, T$ : a weakly stationary series observed over  $T$  periods
  - ▶  $\mu = Ex_t$ : its mean;  $\gamma_j = \text{Cov}(x_t, x_{t-j})$ : the autocovariance at lag  $j$ , the same for every  $t$
  - ▶  $\gamma_0 = \text{Var}(x_t)$ ;  $\rho_j = \gamma_j/\gamma_0 \in [-1, 1]$ : the autocorrelation at lag  $j$
- Sample counterparts (a hat marks an estimated value)
  - ▶ sample mean  $\bar{x} = T^{-1} \sum_{t=1}^T x_t$
  - ▶ sample autocovariance  $\hat{\gamma}_j = T^{-1} \sum_{t=j+1}^T (x_t - \bar{x})(x_{t-j} - \bar{x})$
  - ▶  $\hat{\gamma}_j$  averages the products of deviations from the mean that lie  $j$  periods apart
- Long-run variance  $\Omega = \sum_{j=-\infty}^{\infty} \gamma_j = 2\pi f(0)$ 
  - ▶ the sum of all autocovariances ( $\gamma_{-j} = \gamma_j$ ); it governs the precision of  $\bar{x}$  (next section)
  - ▶  $f(\omega)$ : the spectral density at frequency  $\omega$ ;  $f(0)$  measures the variance of the slow, long-run movements

## Notation (2/2)

- Kernel  $k(\cdot)$ : a weight function with  $k(0) = 1$ ; lag  $j$  receives the weight  $k(j/S)$ 
  - ▶ bandwidth  $S$ : the scale of the weights; lags much longer than  $S$  get (almost) zero weight
  - ▶ Newey–West:  $L = S - 1$  lags for the Bartlett kernel;  $b = S/T$ : the bandwidth as a share of the sample
- Size of a test: the probability of rejecting a true  $H_0$ 
  - ▶ nominal level 5% unless stated; a test over-rejects when its actual size exceeds the nominal level
- HAC: heteroskedasticity and autocorrelation consistent (an estimator of  $\Omega$ )
  - ▶ HAR: heteroskedasticity and autocorrelation robust (the test built on a HAC estimator)

## Prior knowledge from TSA (bachelor)

| Topic                         | TSA                             | Used in ATS for                                                                 |
|-------------------------------|---------------------------------|---------------------------------------------------------------------------------|
| Stationarity, ACF, Wold       | <a href="#">TSA, Chapter 1</a>  | long-run variance, HAC, bootstrap (this chapter)                                |
| ARMA                          | <a href="#">TSA, Chapter 2</a>  | benchmarks, Monte Carlo designs, sieve bootstrap                                |
| Unit roots, ARIMA             | <a href="#">TSA, Chapter 3</a>  | nonstandard asymptotics, spurious regression, <a href="#">Chapters 2, 4, 16</a> |
| GARCH                         | <a href="#">TSA, Chapter 5</a>  | martingale differences, <a href="#">Chapters 8–9</a>                            |
| VAR, Granger                  | <a href="#">TSA, Chapter 6</a>  | structural VAR, local projections, BVAR ( <a href="#">Chapters 3, 5</a> )       |
| Cointegration, VECM           | <a href="#">TSA, Chapter 7</a>  | ARDL, panel cointegration ( <a href="#">Chapter 4</a> )                         |
| Long memory                   | <a href="#">TSA, Chapter 8</a>  | local Whittle, rough volatility ( <a href="#">Chapter 10</a> )                  |
| State space, Markov switching | <a href="#">TSA, Chapter 10</a> | filtering and regimes ( <a href="#">Chapters 6, 7</a> )                         |

- Slides of the bachelor course: [TSA, Chapter 1](#) to [TSA, Chapter 10](#); each ATS chapter links to the TSA chapter it extends

## Stationarity and the Wold representation

- TSA, Chapter 1. Weak stationarity: constant mean and  $\gamma_j$  depending only on  $j$ 
  - ▶ strict stationarity: all finite-dimensional distributions are invariant to a shift in time
  - ▶ neither implies the other without moment conditions: Cauchy i.i.d. is strictly but not weakly stationary (no variance)
- Wold theorem Hamilton (1994): every weakly stationary, purely nondeterministic series is a weighted sum of past innovations
  - ▶  $x_t - \mu = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ ,  $\psi_0 = 1$ ,  $\sum_j \psi_j^2 < \infty$
  - ▶  $\varepsilon_t$ : white noise with variance  $\sigma^2$ , the one-step forecast error;  $\psi_j$ : the effect of a shock after  $j$  periods
  - ▶  $\varepsilon_t$  is uncorrelated, not necessarily independent: GARCH innovations are Wold innovations
- Hence white noise is not enough for i.i.d.-based inference on squares or on extremes
  - ▶ In this chapter:  $\Omega = \sigma^2 (\sum_j \psi_j)^2$ ;  $\sum_j \psi_j$  is the cumulative (long-run) effect of a shock, so  $\Omega$  is the variance of the permanent component

## ARMA and unit roots (1/2)

- TSA, Chapter 2. ARMA( $p, q$ ):  $\phi(L)x_t = \theta(L)\varepsilon_t$ 
  - ▶  $L$ : the lag operator,  $Lx_t = x_{t-1}$ ;  $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ ;  $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$
  - ▶ stationary if the roots of  $\phi(z)$  lie outside the unit circle; invertible if those of  $\theta(z)$  do
- Long-run variance in closed form:  $\Omega = \sigma^2 \theta(1)^2 / \phi(1)^2$ 
  - ▶  $\phi(1) = 1 - \sum_i \phi_i$  and  $\theta(1) = 1 + \sum_i \theta_i$ : the two polynomials evaluated at  $z = 1$
  - ▶ AR(1):  $\Omega = \sigma^2 / (1 - \phi)^2$ ; the closer  $\phi(1)$  is to 0 (more persistence), the larger  $\Omega$

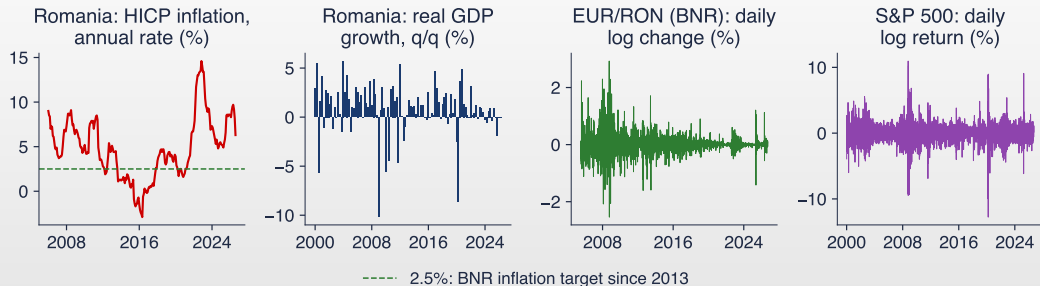
## ARMA and unit roots (2/2)

- TSA, Chapter 3. Unit root,  $\phi(1) = 0$ :  $\Omega = \infty$  and the mean is not defined
  - ▶ the  $t$ -statistic has a nonstandard (Dickey–Fuller) limit
- Spurious regression: one random walk regressed on an independent one
  - ▶  $|t| > 2$  in 76 of 100 regressions in Granger and Newbold (1974)
  - ▶ the  $t$ -statistic diverges at rate  $\sqrt{T}$  Phillips (1986)
  - ▶ no HAC estimator repairs a spurious regression (Seminar 0, B5)
- Near unit roots ( $\phi$  close to 1) are the hard case of this chapter
  - ▶ every HAC method over-rejects there (Monte Carlo section)

## GARCH, VAR, cointegration, state space

- GARCH (TSA, Chapter 5):  $r_t = \sigma_t z_t$  is a martingale difference;  $r_t^2$  is an ARMA(1,1) with strong, slowly decaying autocorrelation
  - ▶  $r_t$ : the return;  $\sigma_t$ : its conditional volatility;  $z_t$ : i.i.d. with mean 0 and variance 1
  - ▶ Means of returns: HAC barely matters; means of squared returns (variances, risk premia): HAC matters a lot
- VAR (TSA, Chapter 6): Granger causality tests are Wald tests; with heteroskedastic errors they need robust covariances
  - ▶ Local projections (Chapter 3) have serially correlated errors by construction: HAC is part of the method
- Cointegration (TSA, Chapter 7): long-run relations; the long-run variance reappears in FM-OLS and in the KPSS statistic
- State space (TSA, Chapter 10): the Kalman filter gives the Gaussian likelihood; its standard errors still need the sandwich form under misspecification

## The series of this chapter



- Romanian HICP inflation, January 2006 – August 2026: from  $-2.9\%$  (May 2016) to  $14.6\%$  (November 2022); last value  $6.3\%$
- Romanian GDP growth: 106 quarters, minimum  $-10.2\%$  in 2009Q1; daily: 5 352 EUR/RON and 6 714 S&P 500 observations

## Interpreting the four series

- ▣ Inflation moves in long swings: years above and years below the 2.5% line
  - ▶ Each monthly value shares eleven months with the previous one (annual rate): overlap adds to persistence
- ▣ GDP growth looks close to white noise, apart from two crisis quarters
  - ▶ With  $T = 106$ , two outliers dominate the sample variance
- ▣ EUR/RON and the S&P 500: volatility clusters (calm years, crisis bursts): dependence in the second moment
- ▣ What do you think? Which of the four series has the largest long-run variance relative to its variance?
  - ▶ Answer: inflation, the most persistent series; the ratios are on the slide “Interpreting the autocorrelations”

## Autocorrelation in six series



- Sample ACF, lags 1–24, with the  $\pm 1.96/\sqrt{T}$  band of the i.i.d. hypothesis (shaded)
- $\hat{\rho}_1$ : S&P 500  $-0.10$ , squared  $0.31$ , EUR/RON  $0.17$ , BET  $0.11$ , GDP  $-0.03$ , inflation  $0.98$

## Interpreting the autocorrelations

- ▣ S&P 500 returns: small negative  $\hat{\rho}_1$ ; squared returns: all 24 lags above the band,  $\hat{\rho}_{12} = 0.22$ 
  - ▶ Long-run variance over variance (Andrews bandwidth): 0.8 for returns, 5.7 for squared returns
- ▣ BET and EUR/RON: positive  $\hat{\rho}_1$  (thin trading, central-bank smoothing of the reference rate)
  - ▶ Ratios 1.2 and 1.1: modest, but enough to move a borderline  $t$ -statistic
- ▣ Inflation:  $\hat{\rho}_1 = 0.98$  and a ratio of 27.0: a naive standard error is about five times too small
- ▣ The rest of the chapter: what these ratios mean, how to estimate them, and when the estimates fail

## Recap: the refresher map

- TSA gave the models; ATS needs the inference that goes with dependent data
- White noise is not independence: GARCH returns are uncorrelated, their squares are not
- The long-run variance  $\Omega = \sigma^2\theta(1)^2/\phi(1)^2$  links ARMA models to the precision of a sample mean
- In our data the dependence ranges from negligible (GDP growth) to extreme (inflation, squared returns)

## The variance of a mean under dependence

- Exact, for any weakly stationary series: the variance of the scaled mean adds up the covariances of all pairs of observations
  - ▶  $\text{Var}(\sqrt{T} \bar{x}) = \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \gamma_{t-s} = \sum_{|j| < T} \left(1 - \frac{|j|}{T}\right) \gamma_j$
  - ▶  $\gamma_{t-s}$ : the covariance of  $x_t$  and  $x_s$ ;  $T - |j|$  pairs lie at distance  $j$ , hence the weight  $1 - |j|/T$
- If  $\sum_j |\gamma_j| < \infty$ , the weights tend to 1 (dominated convergence):
  - ▶  $\text{Var}(\sqrt{T} \bar{x}) \rightarrow \Omega = \gamma_0 + 2 \sum_{j \geq 1} \gamma_j = 2\pi f(0)$
- The i.i.d. formula uses  $\gamma_0$  only; the error factor is  $\Omega/\gamma_0 = 1 + 2 \sum_{j \geq 1} \rho_j$ 
  - ▶ factor above 1: the naive standard error is too small; below 1: too large
  - ▶ effective sample size  $T\gamma_0/\Omega$ : the number of independent observations with the same precision
  - ▶ AR(1) with  $\phi = 0.9$ : 1000 observations are worth about 53 independent ones
- Two questions for the rest of the section
  - ▶ LLN: when does  $\bar{x} \rightarrow \mu$ ?
  - ▶ CLT: when is  $\sqrt{T}(\bar{x} - \mu)/\sqrt{\Omega}$  asymptotically  $N(0, 1)$ ?

## Ergodicity

- A strictly stationary  $\{x_t\}$  is **ergodic** if every shift-invariant event (unchanged when the whole path is shifted in time) has probability 0 or 1
  - ▶ Intuition: one long path visits the whole distribution; time averages equal ensemble averages
- **Ergodic theorem Birkhoff (1931)**: stationary, ergodic,  $E|x_t| < \infty \Rightarrow \bar{x} \rightarrow Ex_t$  almost surely
  - ▶ Functions of finitely many lags of an ergodic process are ergodic: sample moments, autocovariances, OLS
- Counterexample:  $x_t = Z + \varepsilon_t$ , with  $Z$  a random level drawn once and  $\varepsilon_t$  i.i.d.: stationary, not ergodic,  $\bar{x} \rightarrow Z$ 
  - ▶ A single history of one economy is a single draw of  $Z$ : why regime and break questions matter ([Chapters 2, 7](#))



George David Birkhoff (1884–1944)

Photo: unknown author (c. 1910); public domain; [Wikimedia Commons](#)

## Mixing: dependence that fades (1/2)

▣ Strong ( $\alpha$ -) mixing coefficient [Rosenblatt \(1956\)](#):

- ▶  $\alpha(m) = \sup_t \sup\{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{F}_{-\infty}^t, B \in \mathcal{F}_{t+m}^\infty\}$
- ▶  $\mathcal{F}_{-\infty}^t$ : the events determined by  $x_s, s \leq t$  (the past);  $\mathcal{F}_{t+m}^\infty$ : those determined by  $x_s, s \geq t+m$  (the future after a gap of  $m$  periods)
- ▶  $|P(A \cap B) - P(A)P(B)|$  is 0 when  $A$  and  $B$  are independent:  $\alpha(m)$  is the largest dependence between a past and a future event  $m$  periods apart

▣  $\alpha$ -mixing:  $\alpha(m) \rightarrow 0$  as  $m \rightarrow \infty$ , i.e. the distant past and future become independent

- ▶  $\beta(m) = \sup_t E[\sup_{B \in \mathcal{F}_{t+m}^\infty} |P(B | \mathcal{F}_{-\infty}^t) - P(B)|]$ : the average gap between conditional and unconditional probabilities of future events
- ▶  $\varphi(m) = \sup_t \sup\{|P(B | A) - P(B)| : A \in \mathcal{F}_{-\infty}^t, P(A) > 0, B \in \mathcal{F}_{t+m}^\infty\}$ : the worst case over past events
- ▶  $\alpha(m) \leq \beta(m) \leq \varphi(m)$ :  $\beta$ - and  $\varphi$ -mixing are stronger; mixing implies ergodicity [Bradley \(2005\)](#)

## Mixing: dependence that fades (2/2)

- Models that are mixing
  - ▶ stationary ARMA with continuous innovations: geometrically  $\beta$ -mixing [Mokkadem \(1988\)](#)
  - ▶ strictly stationary GARCH(1,1), stochastic volatility: geometrically  $\beta$ -mixing [Carrasco and Chen \(2002\)](#)
  - ▶ geometric:  $\beta(m) \leq C\lambda^m$  for some constants  $C > 0$  and  $0 < \lambda < 1$
- Models that are not
  - ▶ unit roots, long memory with  $d > 0$  ([TSA, Chapter 8](#)), some deterministic chaos
  - ▶ there mixing fails, or is too slow for the standard CLT
- Mixing coefficients cannot be estimated from one path
  - ▶ the assumption is checked through the model, not tested

## Laws of large numbers for dependent data

- **$L^2$  law**: weakly stationary with  $\gamma_j \rightarrow 0 \Rightarrow E(\bar{x} - \mu)^2 \rightarrow 0$ 
  - ▶ Proof:  $E(\bar{x} - \mu)^2 = T^{-1} \sum_{|j| < T} (1 - |j|/T) \gamma_j$ , and the Cesàro mean (the average of the first terms) of the sequence  $\gamma_j \rightarrow 0$  tends to 0
- **Ergodic law Birkhoff (1931)**: almost sure convergence, only  $E|x_t| < \infty$ , no second moment
  - ▶ Covers squared returns of a GARCH with infinite fourth moment: means converge, but the CLT may fail
- **Mixingales and near-epoch dependence Hamilton (1994)**
  - ▶ mixingale:  $\|E(x_t - \mu | \mathcal{F}_{t-m})\|_2 \leq c_t \psi_m$  with  $\psi_m \rightarrow 0$ : the past predicts less and less of  $x_t$
  - ▶ near-epoch dependence:  $\|x_t - E(x_t | v_{t-m}, \dots, v_{t+m})\|_2 \rightarrow 0$  as  $m \rightarrow \infty$ , with  $\{v_t\}$  mixing:  $x_t$  is almost a function of nearby values of a mixing process
  - ▶  $\|z\|_2 = (Ez^2)^{1/2}$ ;  $c_t$ : scaling constants; these cover functions of mixing processes, e.g. the squares of a GARCH
- Consistency (LLN) is cheap; valid standard errors (CLT plus a consistent  $\hat{\Omega}$ ) are the hard part

## Martingale differences and their central limit theorem

- $\{u_t\}$  is a **martingale difference sequence** (MDS) if  $E(u_t \mid \mathcal{F}_{t-1}) = 0$ 
  - ▶  $\mathcal{F}_{t-1}$ : the information available at  $t - 1$ ; the past does not predict  $u_t$  on average
  - ▶ Uncorrelated with every function of the past, but may be conditionally heteroskedastic: GARCH returns, scores of a correct likelihood, one-step forecast errors of an optimal forecast
- **MDS CLT Billingsley (1961)**: stationary, ergodic MDS with  $E u_t^2 = \sigma^2 < \infty \Rightarrow \sqrt{T} \bar{u} \rightarrow_d N(0, \sigma^2)$ 
  - ▶  $\rightarrow_d$ : convergence in distribution; nonstationary MDS: Lindeberg-type conditions [Brown \(1971\)](#)
  - ▶ Here  $\Omega = \gamma_0$ : no autocorrelation correction is needed, only heteroskedasticity-robust errors [White \(1980\)](#)
- Overlapping  $h$ -step forecast errors are not an MDS: they follow an  $MA(h - 1)$  (Section “Overlapping observations”)

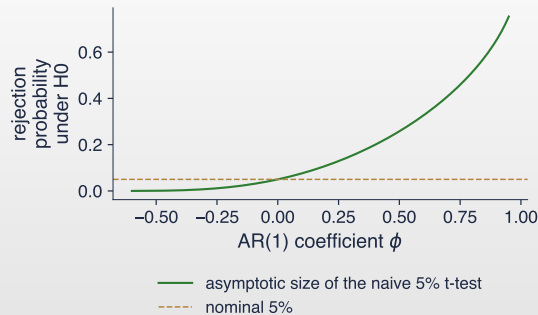
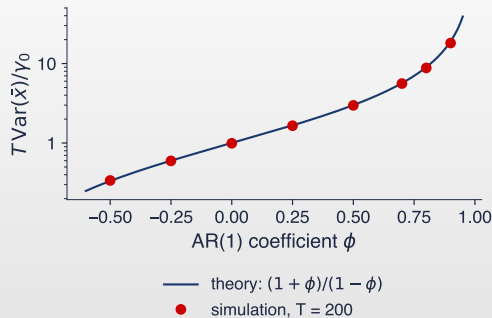
## Central limit theorems beyond martingale differences (1/2)

- ▣ **Mixing CLT Ibragimov (1962)**: if  $x_t$  is stationary and
  - ▶  $E|x_t|^{2+\delta} < \infty$ ,  $\sum_m \alpha(m)^{\delta/(2+\delta)} < \infty$ ,  $\Omega > 0 \Rightarrow \sqrt{T}(\bar{x} - \mu) \rightarrow_d N(0, \Omega)$
  - ▶  $\delta > 0$ : the number of moments beyond the second;  $\alpha(m)$ : the mixing coefficient of the previous section
  - ▶ trade-off: more moments (larger  $\delta$ ) allow slower mixing; the first CLT under strong mixing is [Rosenblatt \(1956\)](#)
- ▣ **Linear processes Phillips and Solo (1992)**:  $x_t - \mu = \sum \psi_j \varepsilon_{t-j}$  with i.i.d. or MDS  $\varepsilon_t$  and  $\sum j|\psi_j| < \infty$ 
  - ▶ the condition on  $\psi_j$  requires the effect of a shock to die out fast enough

## Central limit theorems beyond martingale differences (2/2)

- ▣ Beveridge–Nelson decomposition of the lag polynomial  $\psi(L) = \sum_j \psi_j L^j$ :
  - ▶  $\psi(L) = \psi(1) - (1 - L)\tilde{\psi}(L)$ , so  $\sqrt{T}(\bar{x} - \mu) = \psi(1)\sqrt{T}\bar{\varepsilon} + o_p(1)$
  - ▶  $\psi(1) = \sum_j \psi_j$ : the long-run effect of a shock;  $\tilde{\psi}(L)$ : the polynomial of the transitory part;  $o_p(1)$ : a term that tends to 0 in probability
  - ▶ the mean of  $x_t$  behaves like  $\psi(1)$  times the mean of the innovations, hence  $\Omega = \sigma^2 \psi(1)^2$
- ▣ Functional versions (invariance principles) give the Brownian-motion limits
  - ▶ they underlie the unit-root theory and the fixed- $b$  theory (next section)
- ▣ All three CLTs need  $0 < \Omega < \infty$ 
  - ▶ unit roots give  $\Omega = \infty$ ; over-differencing gives  $\Omega = 0$

## Information lost to dependence



- Left:  $T\text{Var}(\bar{x})/\gamma_0$  for an AR(1), theory  $(1 + \phi)/(1 - \phi)$  and 20 000 simulations with  $T = 200$ ; right: asymptotic size of the naive 5% test,  $2[1 - \Phi(1.96/\sqrt{(1 + \phi)/(1 - \phi)})]$ , with  $\Phi$  the standard normal distribution function

## Interpreting the variance inflation

- $\phi = 0.5$ : the factor is 3 (simulated 2.98);  $\phi = 0.9$ : 19 (simulated 18.1, the finite- $T$  factor  $1 - |j|/T$  lowers it)
  - ▶ The naive test rejects a true  $H_0$  with probability 15% at  $\phi = 0.3$ , 26% at 0.5 and 65% at 0.9
- Negative dependence works the other way:  $\phi = -0.5$  gives a factor 0.33, a conservative naive test
  - ▶ The S&P 500 mean return is an example: its HAC standard error is smaller than the naive one
- Confidence intervals inherit the factor: a naive 95% interval covers  $\mu$  only about 74% of the time at  $\phi = 0.5$

## Regression with dependent errors (1/2)

- Model  $y_t = x_t' \beta + u_t$  with  $E(x_t u_t) = 0$ 
  - ▶  $x_t$ : the vector of regressors;  $\beta$ : the coefficients;  $u_t$ : the error, uncorrelated with  $x_t$ ; ' denotes transposition
- Under a CLT for the products  $x_t u_t$ , the OLS estimator is asymptotically normal:
  - ▶  $\sqrt{T}(\hat{\beta} - \beta) \rightarrow_d N(0, Q^{-1} \Omega_{xu} Q^{-1})$ ,  $Q = E(x_t x_t')$ ,  $\Omega_{xu} = \sum_j E(x_t u_t u_{t-j} x_{t-j}')$
  - ▶  $Q$ : the second-moment matrix of the regressors;  $\Omega_{xu}$ : the long-run variance of  $x_t u_t$
  - ▶ the form  $Q^{-1} \Omega_{xu} Q^{-1}$  is called a sandwich:  $\Omega_{xu}$  between two copies of  $Q^{-1}$

## Regression with dependent errors (2/2)

- Scalar slope,  $x_t$  and  $u_t$  independent stationary series: the error factor is
  - ▶  $\Omega_{xu}/(\gamma_{x,0}\gamma_{u,0}) = 1 + 2 \sum_{j \geq 1} \rho_x(j)\rho_u(j)$
  - ▶  $\gamma_{x,0}, \gamma_{u,0}$ : the variances of  $x_t$  and  $u_t$ ;  $\rho_x(j), \rho_u(j)$ : their autocorrelations at lag  $j$
- A large distortion needs both a persistent regressor and a persistent error
  - ▶ if either autocorrelation is zero, the factor is 1; a persistent predictor with persistent errors is the worst case
- The Newey–West covariance matrix [Newey and West \(1987\)](#): the sandwich  $\hat{Q}^{-1}\hat{\Omega}_{xu}\hat{Q}^{-1}$ 
  - ▶  $\hat{Q}$ : the sample mean of  $x_t x_t'$ ;  $\hat{\Omega}_{xu}$ : a kernel estimate of  $\Omega_{xu}$  (next section)

## Recap: asymptotics for dependent data

- ▣  $\text{Var}(\sqrt{T}\bar{x}) \rightarrow \Omega = \sum_j \gamma_j = 2\pi f(0)$ ; the i.i.d. formula keeps only  $\gamma_0$
- ▣ Ergodicity gives the LLN; mixing or martingale structure plus moments give the CLT
- ▣ For an MDS,  $\Omega = \gamma_0$ : heteroskedasticity-robust errors suffice
- ▣ Positive persistence makes naive tests over-reject; negative persistence makes them conservative

## Estimating the long-run variance

- Plugging in all sample autocovariances fails:  $\sum_{|j| < T} \hat{\gamma}_j = 0$  identically for demeaned data
  - ▶ high-order  $\hat{\gamma}_j$  use few pairs and are mostly noise
- **Kernel estimator:** a weighted sum of sample autocovariances
  - ▶  $\hat{\Omega} = \sum_{|j| < T} k(j/S) \hat{\gamma}_j$ ,  $k(0) = 1$ ,  $k(x)$  decreasing in  $|x|$
  - ▶  $k(j/S)$ : the weight of lag  $j$ ;  $S$ : the bandwidth; equivalently a smoothed periodogram at frequency 0,  $\hat{\Omega} = 2\pi \hat{f}(0)$  [Shumway and Stoffer \(2017\)](#)
- Consistency:  $S \rightarrow \infty$  and  $S/T \rightarrow 0$ , under mixing and moment conditions [Andrews \(1991\)](#)
  - ▶ the down-weighted lags create bias, the included ones create variance:  $S$  balances the two
- A variance estimator must be nonnegative: not every kernel guarantees this

## Case study: Newey and West (1987) (1/2)

▣ Newey and West (1987), Econometrica 55(3): the Bartlett kernel  $k(x) = 1 - |x|$  for  $|x| \leq 1$ , with  $L$  lags

►  $\hat{\Omega}_{NW} = \hat{\Gamma}_0 + \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j'), \quad \hat{\Gamma}_j = T^{-1} \sum_{t>j} \hat{h}_t \hat{h}_{t-j}'$

▣ Notation

- $\hat{h}_t = x_t \hat{u}_t$ : regressor times OLS residual  $\hat{u}_t$  (the score of observation  $t$ )
- $\hat{\Gamma}_j$ : the sample autocovariance matrix of  $\hat{h}_t$  at lag  $j$ ;  $\hat{\Gamma}_0$ : its variance matrix
- $1 - j/(L+1)$ : weights that decrease linearly from 1 to  $1/(L+1)$ ; lags beyond  $L$  get weight 0

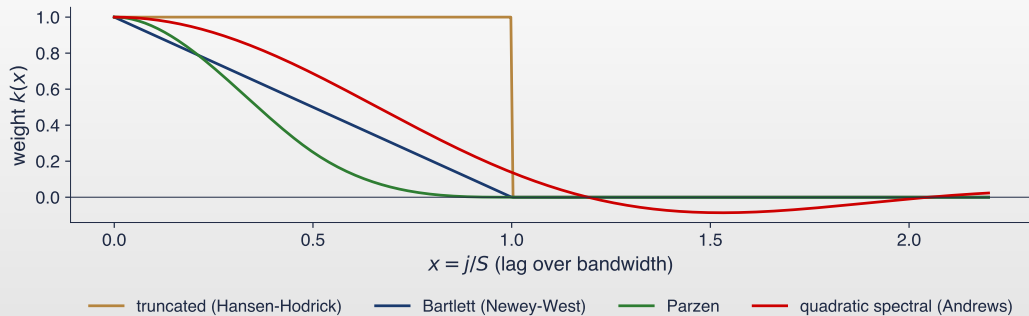
## Case study: Newey and West (1987) (2/2)

- **Positive semi-definite by construction:** Bartlett weights are the autocovariances of a moving sum
  - ▶ up to end effects,  $\hat{\Omega}_{NW} = \frac{1}{(L+1)T} \sum_t \left( \sum_{i=0}^L \hat{h}_{t-i} \right) \left( \sum_{i=0}^L \hat{h}_{t-i} \right)'$
  - ▶ a sum of outer products  $vv'$ , each positive semi-definite, so the sum is too
  - ▶ spectral view: the Bartlett (Fejér) window is nonnegative at every frequency
- Consistency with  $L \rightarrow \infty$ ,  $L = o(T^{1/4})$  in the original paper
  - ▶  $L = o(T^{1/4})$ :  $L$  grows more slowly than  $T^{1/4}$ ; [Andrews \(1991\)](#) refines the rates
- One page of algebra, now the default robust standard error in every econometrics package

## A truncated kernel can give a negative variance

- ▣ Truncated (uniform) kernel:  $k(x) = 1$  for  $|x| \leq 1$ :  $\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^L \hat{\gamma}_j$  (used by Hansen and Hodrick (1980) for overlapping forecasts)
  - ▶ Exactly right when  $\gamma_j = 0$  for  $j > L$ , e.g. an  $\text{MA}(h-1)$  forecast error with  $L = h-1$
- ▣ Ten observations alternating in sign,  $(1, -1, 1, -1, 1, -1, 1, -1, 2, -2)$ :  $\hat{\gamma}_0 = 1.6$ ,  $\hat{\gamma}_1 = -1.3$ ,  $\hat{\gamma}_2 = 1.0$ 
  - ▶ Truncated,  $L = 1$ :  $\hat{\gamma}_0 + 2\hat{\gamma}_1 = -1.0 < 0$
  - ▶ Bartlett,  $L = 1$ :  $\hat{\gamma}_0 + \hat{\gamma}_1 = 0.3 > 0$
- ▣ Lesson: a negative variance is not a numerical accident; it is a property of the kernel (Seminar 0, A3–A4)

## Kernels



- Weight  $k(x)$  against  $x = j/S$ , the lag relative to the bandwidth; Bartlett:  $1 - |x|$  ( $|x| \leq 1$ )
- Parzen:  $1 - 6x^2 + 6|x|^3$  ( $|x| \leq 1/2$ ),  $2(1 - |x|)^3$  ( $1/2 < |x| \leq 1$ ); QS (quadratic spectral):  

$$\frac{25}{12\pi^2 x^2} \left[ \frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5) \right], \text{ all lags}$$

## Interpreting the kernels

- Bartlett, Parzen and QS have nonnegative spectral windows (the Fourier transform of the weights):  $\hat{\Omega} \geq 0$  always
  - ▶ QS takes small negative weights (minimum  $-0.086$ ) yet stays positive semi-definite
- Kernel order  $q$ :  $1 - k(x) \sim c|x|^q$  near 0, with  $c > 0$  a constant; Bartlett  $q = 1$ , Parzen and QS  $q = 2$ 
  - ▶ bias of order  $S^{-q}$ , variance of order  $S/T$ : the optimal  $S$  grows like  $T^{1/(2q+1)}$
  - ▶ i.e.  $T^{1/3}$  for Bartlett and  $T^{1/5}$  for QS
- Andrews (1991): QS minimises the asymptotic MSE among kernels with nonnegative estimates

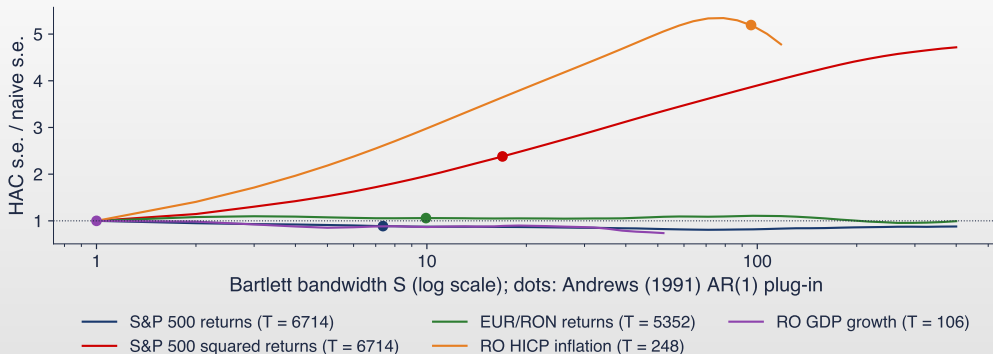
## Case study: Andrews (1991) and the choice of bandwidth (1/2)

- MSE-optimal bandwidth Andrews (1991): the  $S$  that minimises the mean squared error of  $\hat{\Omega}$ 
  - ▶  $S^* = c_k (\alpha(q) T)^{1/(2q+1)}$
  - ▶  $q$ : the order of the kernel;  $c_k$ : a constant of the kernel;  $\alpha(q)$ : a function of the unknown spectrum, large for persistent series
  - ▶ Bartlett:  $S^* = 1.1447(\hat{\alpha}(1) T)^{1/3}$ ; Parzen:  $2.6614(\hat{\alpha}(2) T)^{1/5}$ ; QS:  $1.3221(\hat{\alpha}(2) T)^{1/5}$
- **AR(1) plug-in**: fit an AR(1) with coefficient  $\hat{\rho}$  to  $\hat{h}_t$ , then
  - ▶  $\hat{\alpha}(1) = \frac{4\hat{\rho}^2}{(1 - \hat{\rho})^2(1 + \hat{\rho})^2}, \quad \hat{\alpha}(2) = \frac{4\hat{\rho}^2}{(1 - \hat{\rho})^4}$
  - ▶ both grow without bound as  $\hat{\rho} \rightarrow 1$ : more persistence, wider bandwidth

## Case study: Andrews (1991) and the choice of bandwidth (2/2)

- Newey–West rule of thumb:  $L = \lfloor 4(T/100)^{2/9} \rfloor$ 
  - ▶  $\lfloor \cdot \rfloor$ : the integer part;  $L$  depends only on  $T$ , not on the data
  - ▶ their nonparametric automatic selection Newey and West (1994) estimates the spectrum instead
- Prewhitening with a VAR(1), then recolouring Andrews and Monahan (1992)
  - ▶ filter out the AR(1) part, estimate  $\Omega$  on the residuals, then undo the filter: less bias for persistent series
- The MSE criterion targets  $\hat{\Omega}$ , not the test
  - ▶ this is the source of the size distortions of the next sections

## HAC standard errors as a function of the bandwidth



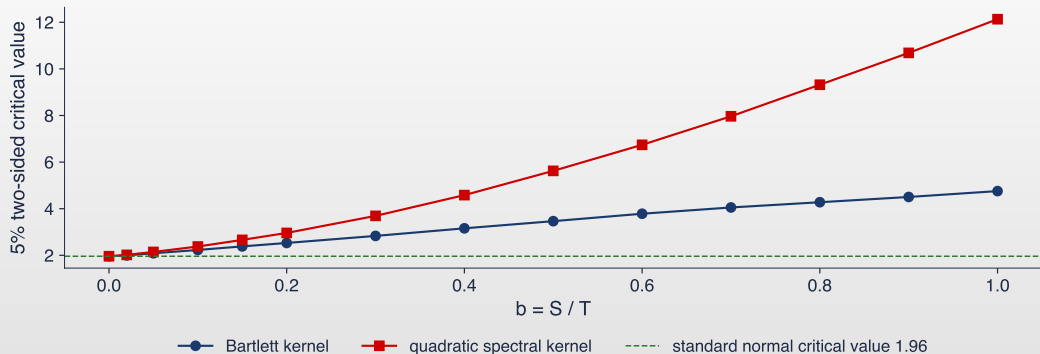
- Ratio of the Bartlett HAC standard error of the mean to the naive one,  $S$  from 1 to  $\min(400, T/2)$ ; dots: Andrews AR(1) plug-in

## Interpreting the bandwidth paths

- Squared S&P 500 returns: ratio 2.04 with the rule of thumb ( $S = 11$ ), 2.38 with Andrews ( $S = 17$ ), still rising to 4.72 at  $S = 400$ 
  - ▶ No plateau: the AR(1) plug-in misses the slow decay of volatility autocorrelation (long memory, [Chapter 10](#))
- Romanian inflation: 2.18 with the rule of thumb, 5.19 with Andrews ( $S = 96$  months)
  - ▶ The rule of thumb ignores the data: with  $T$  fixed it gives the same  $S$  for white noise and for a near unit root
- S&P 500 returns: ratio below 1 at every  $S$  (negative  $\hat{\rho}_1$ ); EUR/RON: flat near 1.06

## Fixed- $b$ asymptotics

- Standard theory:  $S/T \rightarrow 0$ ,  $\hat{\Omega} \rightarrow_p \Omega$ , the  $t$ -statistic is  $N(0, 1)$ 
  - ▶  $\rightarrow_p$ : convergence in probability; the sampling error of  $\hat{\Omega}$  is ignored
  - ▶ with 100–300 observations,  $\hat{\Omega}$  is noisy and biased downwards: the normal critical values are too small
- **Fixed- $b$**  Kiefer, Vogelsang and Bunzel (2000), Kiefer and Vogelsang (2002), Kiefer and Vogelsang (2005): keep  $b = S/T$  fixed as  $T \rightarrow \infty$ 
  - ▶  $\hat{\Omega}/\Omega \rightarrow_d Q_k(b)$ ,  $t \rightarrow_d W(1)/\sqrt{Q_k(b)}$
  - ▶  $W(r)$ : a standard Brownian motion on  $r \in [0, 1]$  ( $r$ : the fraction of the sample);  $Q_k(b)$ : a random functional of the Brownian bridge  $\tilde{W}(r) = W(r) - rW(1)$
  - ▶ Bartlett kernel:  $Q(b) = \frac{2}{b} \int_0^1 \tilde{W}(r)^2 dr - \frac{2}{b} \int_0^{1-b} \tilde{W}(r+b) \tilde{W}(r) dr$  Kiefer and Vogelsang (2005)
  - ▶ nonstandard but pivotal (free of unknown parameters): critical values depend only on the kernel and on  $b$
- Higher-order theory: fixed- $b$  critical values remove the leading size distortion of the normal approximation Sun, Phillips and Jin (2008)

Fixed- $b$  critical values

□ Two-sided 5% critical values of the  $t$ -test for a mean with  $S = bT$ , simulated (20 000 i.i.d. normal samples,  $T = 500$ )

## Interpreting the fixed- $b$ critical values

- Bartlett: 2.23 at  $b = 0.1$ , 3.47 at  $b = 0.5$ , 4.76 at  $b = 1$  (Kiefer–Vogelsang–Bunzel statistic)
  - ▶ Close to the cubic approximation reported in [Kiefer and Vogelsang \(2005\)](#) for the Bartlett kernel
- QS: 2.38 at  $b = 0.1$  and 5.62 at  $b = 0.5$ : QS with a large  $b$  is very noisy
- Larger  $b$ : less bias in  $\hat{\Omega}$ , more variance, compensated by a larger critical value: better size, lower power
- What do you think? Why does the critical value at  $b \rightarrow 0$  tend to 1.96?
  - ▶ Answer: as  $b \rightarrow 0$ ,  $Q_k(b) \rightarrow 1$  ( $\hat{\Omega}$  becomes consistent), so  $t \rightarrow_d W(1) \sim N(0, 1)$ : the standard asymptotics return

## Recommendations for practice: Lazarus, Lewis, Stock and Watson (2018) (1/2)

- Lazarus et al. (2018) choose the bandwidth for the *test*: a size–power frontier, not the MSE of  $\hat{\Omega}$ 
  - ▶ first recommendation: Newey–West with  $S = 1.3\sqrt{T}$  and fixed- $b$  critical values
- Second recommendation: the equal-weighted cosine (EWC) estimator
  - ▶  $\hat{\Omega} = \nu^{-1} \sum_{j=1}^{\nu} \hat{\Lambda}_j^2$ ,  $\hat{\Lambda}_j = \sqrt{2/T} \sum_t \cos[\pi j(t - 1/2)/T] \hat{h}_t$ ,  $\nu = 0.4 T^{2/3}$
  - ▶  $\hat{\Lambda}_j$ : the projection of  $\hat{h}_t$  on the  $j$ -th cosine, a low-frequency component; each  $\hat{\Lambda}_j^2$  estimates  $\Omega$
  - ▶  $\nu$ : the number of cosines averaged, also the degrees of freedom: critical values from Student  $t_\nu$

## Recommendations for practice: Lazarus, Lewis, Stock and Watson (2018) (2/2)

- The size–power trade-off is unavoidable
  - ▶ Lazarus, Lewis and Stock (2021) give the frontier for kernel tests
- Very persistent series ( $\phi$  near 1): no kernel or EWC rule is reliable
  - ▶ Müller (2014) proposes tests valid under local-to-unity dependence (a root that tends to 1 as  $T$  grows)
- These are the defaults of the course from Chapter 1 on

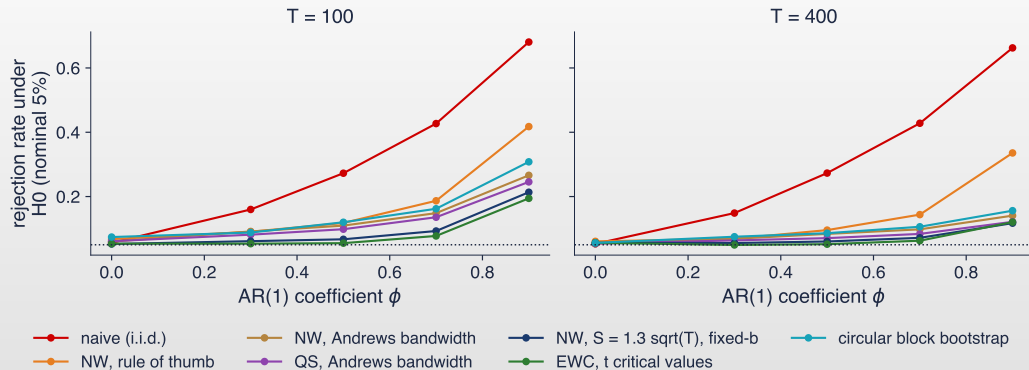
## Recap: HAC estimation

- Kernel estimators down-weight distant lags; Bartlett (Newey–West), Parzen and QS are always nonnegative; the truncated kernel is not
- The [Andrews \(1991\)](#) bandwidths are MSE-optimal for  $\hat{\Omega}$  and adapt to persistence; the rule of thumb does not
- Fixed- $b$  critical values account for the noise in  $\hat{\Omega}$ ; LLSW: NW with  $S = 1.3\sqrt{T}$  or EWC with  $\nu = 0.4T^{2/3}$

## Monte Carlo design

- DGP:  $x_t = \phi x_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \sim N(0, 1)$  i.i.d., 200 burn-in values;  $H_0 : \mu = 0$  is true
  - ▶  $\phi$ : the AR(1) coefficient, i.e. the persistence; burn-in: initial values dropped so that the series starts from its stationary distribution
  - ▶  $\phi \in \{0, 0.3, 0.5, 0.7, 0.9\}$ ,  $T \in \{100, 400\}$ ; 5 000 replications (1 000 for the bootstrap, 399 resamples each); seed 2026
- Seven tests at nominal 5%
  - ▶ naive i.i.d.; NW rule of thumb; NW and QS with the Andrews AR(1) bandwidth (normal critical values)
  - ▶ NW with  $S = 1.3\sqrt{T}$  and fixed- $b$  values; EWC with  $t_\nu$ ; circular block bootstrap with the Politis–White block length
- Monte Carlo standard error of a rejection rate  $p$  estimated from  $R$  replications:  $\sqrt{p(1-p)/R}$ 
  - ▶ about 0.3 percentage points (pp) at  $p = 5\%$  and  $R = 5000$
- The same design as the Monte Carlo sections of [Andrews \(1991\)](#) and [Kiefer and Vogelsang \(2005\)](#): AR(1) data, rejection rates under the null

## Size of seven tests under AR(1) dependence



Rejection rate of  $H_0 : \mu = 0$  (true) against  $\phi$ ; dotted line: 5%

## Size in numbers (%)

| Test                           | $T = 100$  |      |      | $T = 400$  |      |      |
|--------------------------------|------------|------|------|------------|------|------|
|                                | $\phi = 0$ | 0.5  | 0.9  | $\phi = 0$ | 0.5  | 0.9  |
| naive i.i.d.                   | 5.6        | 27.3 | 68.1 | 5.3        | 27.3 | 66.2 |
| NW, rule of thumb              | 6.8        | 11.8 | 41.8 | 6.1        | 9.6  | 33.6 |
| NW, Andrews                    | 6.0        | 11.0 | 26.6 | 5.3        | 8.4  | 14.1 |
| QS, Andrews                    | 6.1        | 9.8  | 24.6 | 5.4        | 7.1  | 12.1 |
| NW, $1.3\sqrt{T}$ , fixed- $b$ | 5.2        | 6.7  | 21.3 | 5.6        | 6.0  | 11.7 |
| EWC, $t_\nu$                   | 5.3        | 5.5  | 19.4 | 5.6        | 5.2  | 12.1 |
| circular block bootstrap       | 7.4        | 12.0 | 30.8 | 5.6        | 8.6  | 15.6 |

- Monte Carlo standard errors: about 0.3 pp near 5%, about 0.7 pp near 30% (bootstrap rows: about twice as large)

## Interpreting the Monte Carlo

- Naive test: 27.3% at  $\phi = 0.5$  and 68.1% at  $\phi = 0.9$ ; more data do not help (66.2% at  $T = 400$ )
  - ▶ The distortion is a property of the estimator, not of the sample size
- NW with the rule of thumb: 41.8% at  $\phi = 0.9$ ,  $T = 100$ ; Andrews bandwidths halve the excess; fixed- $b$  and EWC come closest (21.3%, 19.4%)
  - ▶ At  $T = 400$  and  $\phi = 0.5$ : 6.0% and 5.2%, essentially exact
- The block bootstrap is not a free lunch: 30.8% at  $\phi = 0.9$ ,  $T = 100$ , between NW and fixed- $b$
- Our inflation series has  $\hat{\rho}_1 = 0.98$ : outside the range where any of these tests is reliable (AI mini-case)

## Recap: size distortions

- Always report the actual size of the procedure you use, at the persistence of your data
- Fixed- $b$  NW and EWC ([Lazarus et al. \(2018\)](#)) dominate the classical choices in size at a modest cost in power
- A Monte Carlo is an experiment: fix the seed, report  $R$  and the Monte Carlo standard error

## The failure of the i.i.d. bootstrap

- Efron's bootstrap [Efron \(1979\)](#) resamples single observations: the dependence is destroyed
  - ▶ a star marks a bootstrap quantity:  $\bar{x}^*$  is the mean of one resample,  $\text{Var}^*$  the variance over resamples
  - ▶  $\text{Var}^*(\sqrt{T}\bar{x}^*) = \hat{\sigma}_0^2$ , not  $\Omega$ : the i.i.d. bootstrap reproduces the naive standard error
- [Singh \(1981\)](#): even for  $m$ -dependent data (independent beyond distance  $m$ ) the i.i.d. bootstrap of the mean is inconsistent
  - ▶ Idea of the block bootstrap: resample blocks long enough to keep the dependence inside them
- Price to pay: dependence across block joints is lost, and the block length becomes a tuning parameter



Bradley Efron (2007)

Photo: [Ryan K. Morris, NSTMF \(2007\)](#);  
public domain; [Wikimedia Commons](#)

## Case study: the moving-block bootstrap of Künsch (1989)

- **Künsch (1989)**: overlapping blocks  $B_i = (x_i, \dots, x_{i+l-1})$  of length  $l$ ,  $i = 1, \dots, T - l + 1$ 
  - ▶ draw  $k = \lceil T/l \rceil$  blocks with replacement ( $\lceil \cdot \rceil$ : rounding up), concatenate, keep the first  $T$  values
- Consistency for smooth functions of means if  $l \rightarrow \infty$ ,  $l/T \rightarrow 0$ 
  - ▶  $\text{Var}^*(\sqrt{T}\bar{x}^*) \approx \sum_{|j| < l} (1 - |j|/l) \hat{\gamma}_j$ : a Bartlett estimate with  $S = l$
  - ▶ Optimal  $l \propto T^{1/3}$  for variance and bias, as for the Bartlett bandwidth Lahiri (2003)
- Edge effect: end observations enter fewer blocks, so  $E^*\bar{x}^* \neq \bar{x}$ 
  - ▶  $E^*$ : the expectation over resamples; centre the bootstrap statistics at  $E^*\bar{x}^*$



Hans Rudolf Künsch (2007)

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de; Wikimedia Commons

## Circular and stationary bootstrap

- **Circular block bootstrap** (Politis and Romano (1992); Lahiri (2003)): wrap the data on a circle,  $x_{T+i} = x_i$ 
  - ▶ Every observation enters  $l$  blocks:  $E^* \bar{x}^* = \bar{x}$  exactly (Seminar 0, A5–A6)
- **Stationary bootstrap** Politis and Romano (1994): block lengths geometric with mean  $1/p$ , uniform starting points, data wrapped on a circle
  - ▶ Equivalently: each step continues the block with probability  $1 - p$  or jumps to a random point with probability  $p$
  - ▶ The resampled series is stationary; the method is less sensitive to the choice of  $1/p$  than MBB to  $l$
- Efficiency: MBB and CBB have a smaller asymptotic MSE for the variance than the stationary bootstrap at their optimal lengths Lahiri (2003)
- The stationary bootstrap is the engine of the Reality Check White (2000) and of the SPA test (Chapter 1)

## Choosing the block length (1/2)

- ▣ Politis and White (2004), corrected by Patton, Politis and White (2009): plug-in estimate of the MSE-optimal mean block length

- ▶  $\hat{b}_{opt} = \left( \frac{2\hat{G}^2}{\hat{D}} \right)^{1/3} T^{1/3}, \quad \hat{G} = \sum_{|k| \leq M} \lambda(k/M) |k| \hat{\gamma}_k, \quad \hat{g} = \sum_{|k| \leq M} \lambda(k/M) \hat{\gamma}_k$

- ▶  $\hat{D}_{SB} = 2\hat{g}^2$  (stationary),  $\hat{D}_{CB} = \frac{4}{3}\hat{g}^2$  (circular)

- ▣ Notation

- ▶  $\hat{G}$ : autocovariances weighted by their lag, the bias term;  $\hat{g}$ : an estimate of  $\Omega$ ;  $\hat{D}$ : the variance term
- ▶  $\lambda$ : the flat-top (trapezoid) kernel;  $M = 2\hat{m}$ : the truncation lag
- ▶  $\hat{m}$ : the first lag after which  $K_T$  consecutive autocorrelations are insignificant

## Choosing the block length (2/2)

- The same structure as the Andrews bandwidth
  - ▶ bias term  $G$  over variance term  $D$ , rate  $T^{1/3}$ : persistent series get longer blocks
- Implemented in the Python package `arch` (`optimal_block_length`)
- Our data
  - ▶ about 145 days for squared S&P 500 returns, 25 months for inflation
  - ▶ 0.8 for GDP growth, i.e. the i.i.d. bootstrap

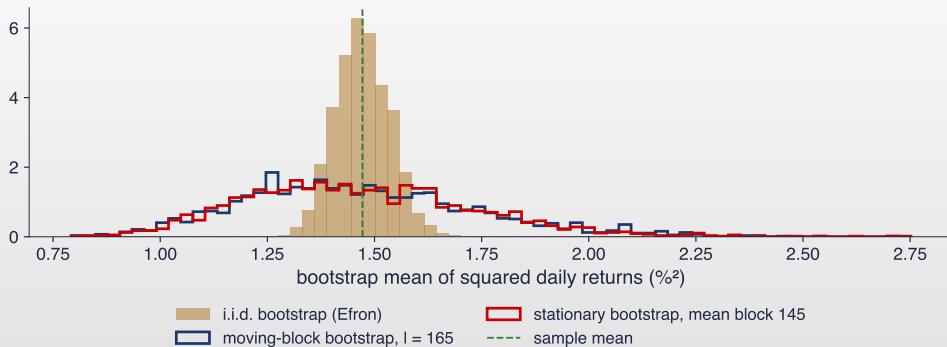
## The wild bootstrap and its limits (1/2)

- ▣ **Wild bootstrap** Wu (1986), Liu (1988), Mammen (1993): each residual is multiplied by a random sign-like weight
  - ▶  $y_t^* = x_t' \hat{\beta} + \hat{u}_t \eta_t$ ,  $\eta_t$  i.i.d.,  $E\eta_t = 0$ ,  $E\eta_t^2 = 1$
  - ▶  $y_t^*$ : the bootstrap observation;  $\eta_t$ : Rademacher ( $\pm 1$  with probability 1/2 each) or the Mammen two-point distribution
- ▣ It keeps the heteroskedasticity of each  $\hat{u}_t$  and destroys every autocorrelation
  - ▶ valid when the scores are an MDS: AR models with GARCH-type errors Gonçalves and Kilian (2004)
  - ▶ **not** valid for overlapping forecasts, local projections or the mean of squared returns

## The wild bootstrap and its limits (2/2)

- Dependent wild bootstrap [Shao \(2010\)](#):  $\eta_t$  drawn from a stationary process
  - ▶  $\text{Corr}(\eta_t, \eta_s) = a(|t - s|/l)$ , with  $a(\cdot)$  a kernel and  $l$  a bandwidth
  - ▶ wild in form, block-like in effect: neighbouring residuals keep their joint signs
- Sieve bootstrap [Bühlmann \(1997\)](#): fit an  $\text{AR}(p)$  with  $p \rightarrow \infty$ , resample its residuals
  - ▶ efficient for linear processes, fragile for nonlinear ones

## Three bootstraps of one mean



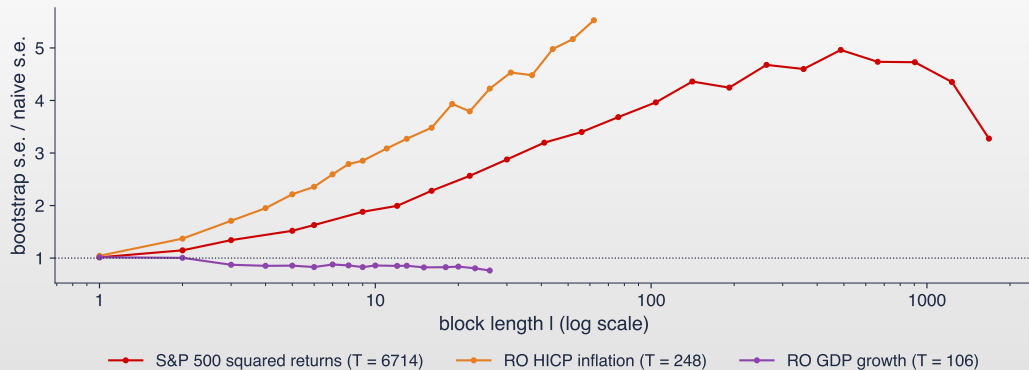
□ Mean of squared daily S&P 500 returns (1.472,  $T = 6714$ ); 1999 resamples each; block lengths from Politis–White

ATS\_ch0\_block\_bootstrap

## Interpreting the three bootstraps

- i.i.d. bootstrap: standard error 0.063, equal to the naive 0.064, as theory says
  - ▶ Block bootstraps: MBB 0.276, CBB 0.268, stationary 0.269: about 4.2 times the naive value
- HAC with the Andrews bandwidth: 0.152 (about 2.4 times the naive value): smaller than the block bootstraps
  - ▶ Block length 145 against an Andrews bandwidth of 17: two plug-in rules, two different views of the same slowly decaying ACF
- When two valid methods disagree by a factor of two, report both and explain why: here, the volatility of volatility has long memory

## Bootstrap standard error against the block length



□ Circular block bootstrap, 999 resamples per length; ratio to the naive standard error

## Interpreting the block-length paths

- $l = 1$  is the i.i.d. bootstrap (ratio 1); the ratio grows with  $l$  while  $l$  is shorter than the memory of the series
  - ▶ Inflation: ratio 4.2 at the Politis–White length of 25 months
- GDP growth: flat near 1 (and slightly below for long blocks): no dependence to capture
- Very long blocks: few distinct blocks, noisy and downward-biased variance (the bias of  $\hat{\gamma}_j$  at large lags)
- What do you think? Why does the inflation curve keep rising until  $l$  is about a quarter of the sample?
  - ▶ Answer: the autocorrelation of inflation decays very slowly (annual rates overlap,  $\hat{\rho}_1 = 0.98$ ); only very long blocks capture it before the downward bias takes over

## Means of six series: standard errors by method

| Series          | $T$   | Mean  | $\hat{\rho}_1$ | naive  | NW-A   | stat. boot. | $t_{\text{naive}} / t_{\text{EWC}}$ |
|-----------------|-------|-------|----------------|--------|--------|-------------|-------------------------------------|
| S&P 500 returns | 6 714 | 0.025 | -0.10          | 0.0148 | 0.0131 | 0.0129      | 1.67 / 2.07                         |
| S&P 500 squared | 6 714 | 1.472 | 0.31           | 0.064  | 0.152  | 0.292       | 23.06 / 6.71                        |
| BET returns     | 6 670 | 0.064 | 0.11           | 0.0171 | 0.0188 | 0.0187      | 3.73 / 2.89                         |
| EUR/RON         | 5 352 | 0.007 | 0.17           | 0.0038 | 0.0040 | 0.0039      | 1.86 / 1.73                         |
| RO GDP growth   | 106   | 0.829 | -0.03          | 0.240  | 0.240  | 0.240       | 3.46 / 3.67                         |
| RO inflation    | 248   | 4.799 | 0.98           | 0.218  | 1.131  | 0.957       | 22.05 / 5.92                        |

- NW-A: Newey–West with the Andrews bandwidth; stationary bootstrap with the Politis–White mean block; returns and changes in % per day, GDP in % per quarter, inflation in %

## Interpreting the table

- S&P 500:  $t$  rises from 1.67 (naive) to 2.03 (fixed- $b$ , critical value 1.99) and 2.07 (EWC): the mean return (6.2% a year) becomes borderline significant
  - ▶ Negative autocorrelation shrinks the standard error: robust inference is not always more conservative
- EUR/RON: mean depreciation of the leu not significant at 5% by any method ( $t$  between 1.73 and 1.86)
- Inflation:  $t$  falls from 22.05 to 5.92 for  $H_0 : \mu = 0$ ; standard errors grow by a factor of about 4–5
- GDP growth: all methods agree; the question there is the two crisis quarters, not the dependence (Seminar 0, B2)

## Recap: the bootstrap for dependent data

- The i.i.d. bootstrap reproduces the naive standard error; resample blocks instead
- MBB [Künsch \(1989\)](#) is a Bartlett HAC estimator in disguise; CBB removes the edge bias; the stationary bootstrap keeps stationarity
- Block length  $\propto T^{1/3}$ , chosen by Politis–White; the wild bootstrap handles heteroskedasticity only

## Case study: Estrella and Hardouvelis (1991) (1/2)

▣ Estrella and Hardouvelis (1991): does the slope of the yield curve predict US real growth?

▶ regression of the average growth over the next  $h$  quarters on the term spread

$$\text{▶ } y_t^{(h)} = \frac{400}{h} \ln \frac{\text{GDP}_{t+h}}{\text{GDP}_t} = \beta_0 + \beta_1(i_t^{10y} - i_t^{3m}) + u_t, \quad h = 4$$

▣ Notation

▶  $y_t^{(h)}$ : real GDP growth from quarter  $t$  to  $t + h$ , annualised, in %; the factor  $400/h$  turns a log change over  $h$  quarters into % a year

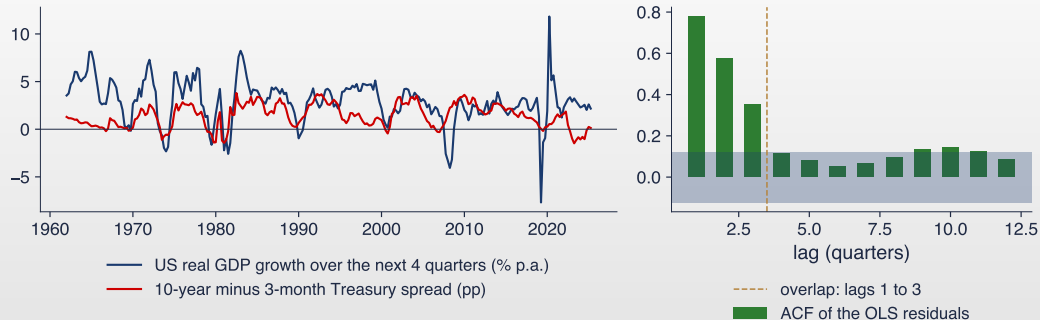
▶  $i_t^{10y}$ ,  $i_t^{3m}$ : the 10-year and 3-month Treasury yields; their difference is the term spread

▶  $\beta_1 > 0$ : a steeper curve announces faster growth;  $u_t$ : the forecast error

## Case study: Estrella and Hardouvelis (1991) (2/2)

- Overlap:  $y_t^{(4)}$  and  $y_{t+1}^{(4)}$  share three quarters
  - ▶ so  $u_t$  is at least MA(3) under  $H_0$ , even if quarterly shocks are independent
  - ▶ classical remedy: Hansen and Hodrick (1980) (truncated kernel,  $h - 1$  lags) or Newey–West with  $L = h - 1$
- Our data: FRED real GDP, 10-year and 3-month Treasury yields (quarterly averages)
  - ▶ 1962Q1 – 2025Q2,  $T = 254$
- Same design for local projections (Chapter 3) and for long-horizon return regressions (MFM)

## The spread and future growth



Left: the two series; right: ACF of the OLS residuals,  $\hat{\rho}_1 = 0.78$ ,  $\hat{\rho}_3 = 0.35$ ,  $\hat{\rho}_4 = 0.11$

ATS\_ch0\_term\_spread

## One slope, six standard errors

| Standard error                          | $\widehat{se}(\hat{\beta}_1)$ | $t$  | Critical value |
|-----------------------------------------|-------------------------------|------|----------------|
| Classical OLS                           | 0.11                          | 4.72 | 1.96           |
| White (heteroskedasticity only)         | 0.10                          | –    | 1.96           |
| Hansen–Hodrick, 3 lags                  | 0.20                          | –    | 1.96           |
| Newey–West, 3 lags                      | 0.17                          | 3.10 | 1.96           |
| NW, Andrews ( $S = 14$ )                | 0.23                          | 2.25 | 1.96           |
| NW, $S = 1.3\sqrt{T} = 21$ , fixed- $b$ | 0.25                          | 2.04 | 2.18           |

□  $\hat{\beta}_1 = 0.52$  pp of annual growth per pp of spread,  $R^2 = 0.08$

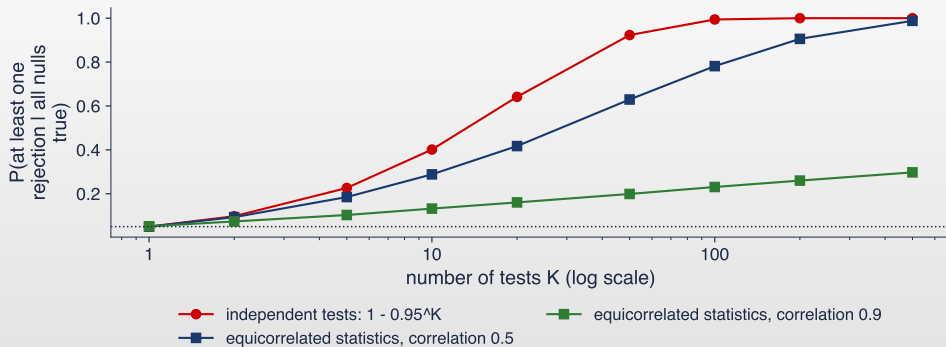
## Interpreting the term-spread regression

- The residuals are more persistent than the MA(3) implied by overlap ( $\hat{\rho}_4 = 0.11$ ): 3 lags are not enough
  - ▶ The standard error doubles from classical OLS to fixed- $b$ :  $t$  from 4.72 to 2.04, just below the critical value 2.18
- Subsamples (NW, 3 lags): 1962–1988  $\hat{\beta}_1 = 1.12$  ( $t = 4.6$ ); 1989–2025  $\hat{\beta}_1 = 0.15$  ( $t = 0.9$ )
  - ▶ The predictive power of the spread weakened after the original sample: a break question ([Chapter 2](#)) and an out-of-sample question ([Chapter 1](#))
- Robust inference first, then stability: a full-sample  $t$  of 4.72 hides both problems

## Data snooping

- **Data snooping:** reusing one data set to select a model, a rule or a specification, then testing it on the same data as if it were chosen in advance
  - ▶ [Lo and MacKinlay \(1990\)](#): sorting portfolios on characteristics already known to be related to returns biases asset-pricing tests
  - ▶ [Harvey, Liu and Zhu \(2016\)](#): more than 300 published return factors; a new factor needs  $t > 3$ , not  $t > 2$
- Error rates for  $K$  tests
  - ▶ FWER (family-wise error rate): probability of at least one false rejection among the  $K$  tests at level  $\alpha$
  - ▶ Bonferroni tests each hypothesis at  $\alpha/K$ ; Holm (step-down; [Holm \(1979\)](#)) compares the ordered p-values  $p_{(i)}$  with  $\alpha/(K - i + 1)$ ; both control the FWER
  - ▶ FDR (false discovery rate): expected share of false rejections among rejections, controlled by [Benjamini and Hochberg \(1995\)](#)
- In forecasting: many models, many horizons, many samples
  - ▶ the honest  $p$ -value is the one of the *search*, not of the winner

## The best of $K$ tests



- Probability that the largest  $|t|$  of  $K$  tests exceeds 1.96 when all nulls are true: independent and equicorrelated statistics (20 000 simulations)

## Interpreting the family-wise error

- ▣ Independent tests: FWER 64% with  $K = 20$  and 99.4% with  $K = 100$ 
  - ▶ Bonferroni would test each at  $0.05/20 = 0.0025$
- ▣ Correlated statistics (similar rules, overlapping samples): FWER 42% ( $K = 20$ ) and 78% ( $K = 100$ ) at correlation 0.5; 23% at 0.9
  - ▶ Bonferroni ignores the correlation and is conservative: the bootstrap of the maximum (next slide) uses it
- ▣ What do you think? With 50 moving-average rules, how many “significant” rules do you expect by chance at 5%?
  - ▶ Answer:  $50 \times 0.05 = 2.5$  if the rules were independent; correlated rules give fewer distinct rejections, but the best  $t$  is still inflated

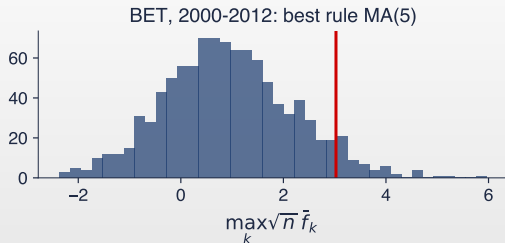
## Case study: the Reality Check of White (2000) (1/2)

- $K$  models against a benchmark, evaluated on  $n$  periods White (2000)
  - ▶  $f_{k,t}$ : the performance difference of model  $k$  at  $t$  (e.g. return of rule  $k$  minus buy-and-hold);  $\bar{f}_k$ : its mean over the  $n$  periods
- Hypothesis and statistic
  - ▶  $H_0 : \max_k E f_{k,t} \leq 0$  (no model beats the benchmark);  $\bar{V} = \max_k \sqrt{n} \bar{f}_k$
  - ▶  $\bar{V}$ : the scaled average outperformance of the best model; a large  $\bar{V}$  is evidence against  $H_0$

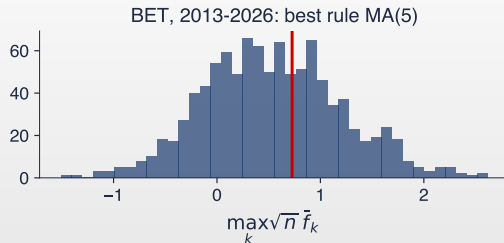
## Case study: the Reality Check of White (2000) (2/2)

- Null distribution by the stationary bootstrap of the whole vector  $f_t$  (keeps the correlation across rules and over time):
  - ▶  $\bar{V}_b^* = \max_k \sqrt{n}(\bar{f}_{k,b}^* - \bar{f}_k)$ ,  $p = B^{-1} \sum_b 1\{\bar{V}_b^* > \bar{V}\}$
  - ▶  $b = 1, \dots, B$ : the resamples;  $\bar{f}_{k,b}^*$ : the mean of rule  $k$  in resample  $b$ ;  $1\{\cdot\}$ : 1 if the condition holds, 0 otherwise
  - ▶  $p$ : the share of resamples with a larger maximum than the observed one, i.e. a  $p$ -value for the whole search
- Sullivan, Timmermann and White (1999) apply it to 7846 technical rules on the Dow Jones; Hansen (2005) (SPA) and Romano and Wolf (2005) (StepM) refine it: Chapter 1
- Our application: 50 rules “long if the close is above its  $n$ -day moving average”,  $n = 5, 10, \dots, 250$ , on the BET
  - ▶ against buy-and-hold, no costs; mean block of 10 days

## The Reality Check on the BET



■ bootstrap distribution of  $V^*$  ( $H_0$ )  
 — observed  $V = 3.03$ ; Reality Check  $p = 0.06$



■ bootstrap distribution of  $V^*$  ( $H_0$ )  
 — observed  $V = 0.73$ ; Reality Check  $p = 0.38$

□ Bootstrap distribution of  $\tilde{V}^*$  under  $H_0$  (999 stationary-bootstrap resamples) and the observed  $\tilde{V}$ ; two subsamples

Q ATS\_ch0\_data\_snooping

## Interpreting the Reality Check

- 2001–2012 ( $n = 2\,993$  days): the best rule MA(5) beats buy-and-hold by 0.055% a day,  $t = 2.51$ , naive one-sided  $p = 0.006$ 
  - ▶ Reality Check  $p = 0.06$ : not significant at 5% once the search over 50 rules is accounted for
- 2014–2026: the same rule,  $t = 0.97$ , Reality Check  $p = 0.38$ : whatever existed did not survive
- The short-MA winner fits the positive  $\hat{\rho}_1$  of BET returns (stale prices)
  - ▶ transaction costs would eat the gain
- Pre-register the rule universe and the sample split before looking at the results (Stage 1 of the project)

## Recap: multiple testing

- The  $p$ -value of the best of  $K$  is not the  $p$ -value of a pre-specified test
- FWER (Bonferroni, Holm) or FDR (Benjamini–Hochberg); for correlated forecasts: bootstrap the maximum (Reality Check)
- Forecast comparison tests (Diebold–Mariano, SPA, Model Confidence Set) are the subject of [Chapter 1](#)

## Seeds and Monte Carlo experiments

- ▣ One seed per project, set once: `rng = np.random.default_rng(2026)`; pass `rng` to every function
  - ▶ Parallel runs: `np.random.SeedSequence(2026).spawn(k)` gives independent streams; never reuse a global state
- ▣ Report what makes a simulation checkable
  - ▶ DGP,  $T$ , number of replications  $R$ , burn-in, seed, Monte Carlo standard error  $\sqrt{p(1-p)/R}$
  - ▶ Results must not depend on the seed beyond the Monte Carlo error: rerun with a second seed
- ▣ Every chart and every number of this chapter is produced by one script with seed 2026 (Quantlets/Ch\_00)

## Versioned data

- ▣ Macro data are revised: the GDP you download today is not the GDP a forecaster saw in real time
  - ▶ Real-time vintages: [Croushore and Stark \(2001\)](#) (Philadelphia Fed), ALFRED for FRED series; Eurostat revises national accounts every quarter
- ▣ Save a dated snapshot of every series used, with the query that produced it
  - ▶ Record the download date, the dataset code (e.g. `namq_10_gdp`) and a checksum (`sha256`) of the file
  - ▶ The course market data are saved once and end on 18 September 2026: every student gets the same numbers
- ▣ A forecast evaluated on revised data answers a different question than the one asked in real time

## Replication packages

- What economics journals now require Vilhuber (2020), Christensen and Miguel (2018)
  - ▶ a README with a data availability statement, the order of the scripts and the expected run time
  - ▶ code that regenerates every table and figure from raw data, without manual steps
  - ▶ the computational environment: `requirements.txt` with versions, or a Colab notebook
- For the ATS project (Stage 2)
  - ▶ a GitHub repository: `data/`, `code/`, `output/`, `README`, `AI_USE.md`, `AI_ERRORS.md`
  - ▶ one command or one notebook reproduces every number in the report
- Test: a colleague clones the repository on a clean machine and gets the same numbers

## Pre-registration and the analysis plan

- Stage 1 of the project fixes, before any estimation:
  - ▶ the hypothesis, the sample and its split (estimation, evaluation), the benchmark
  - ▶ the inference method (HAC rule, bootstrap, critical values) and the multiple-testing correction
  - ▶ the list of specifications that will be reported, all of them
- Why: the garden of forking paths turns many small choices into an implicit search
  - ▶ Every deviation from the plan is allowed but reported, with its reason
- An AI assistant makes the search cheaper: AI\_USE.md records the prompts that shaped a decision

## Recap: reproducible research

- ▣ Seed once, pass the generator, report  $R$  and the Monte Carlo error
- ▣ Dated, checksummed data snapshots; real-time vintages when the question is about real-time forecasting
- ▣ A replication package that runs from raw data to every number; a plan fixed before the results

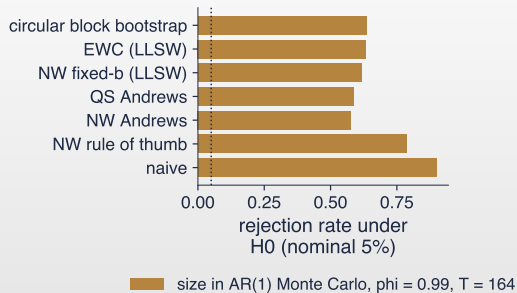
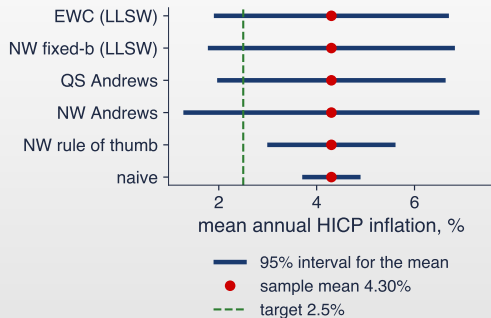
## An open question: valid inference for a near-integrated mean

- Has Romanian inflation averaged the BNR target since 2013? Formally,  $H_0 : \mu = 2.5$  for the HICP annual rate, 164 months to August 2026
  - ▶  $\hat{\rho}_1 = 0.99$ : the series is close to a unit root, and overlapping by construction
- Open: no kernel, EWC or bootstrap rule controls size at this persistence (Monte Carlo above)
  - ▶ Müller (2014) and Lazarus, Lewis and Stock (2021) map the frontier; which test should a central-bank watcher use with 10–15 years of data?
- A question with policy content (credibility of the target) and a methodological core (HAR inference at  $\phi \approx 1$ )

## The AI-assisted discovery loop

- **Literature:** "List peer-reviewed tests for the mean of a highly persistent series since 2010, with DOI and the Monte Carlo design."
  - ▶ Check: resolve every DOI on Crossref; read the size tables yourself (Semantic Scholar, Elicit for search)
- **Hypothesis and code:** "Write Python for NW fixed-b and EWC tests of a mean, and a Monte Carlo of their size at  $\phi = 0.98$ ,  $T = 164$ ."
  - ▶ Check: reproduce a published size number first (e.g. the LLSW tables), then change one thing at a time
- **Robustness and critique:** "Act as a hostile referee: why could this test of the inflation mean be misleading?"
  - ▶ Expect: breaks in the mean (2013, 2022), the CPI-HICP gap, overlap, the choice of start date
- **The human checks:** formulas, timing of information (no look-ahead), every number against the code; the tools are generic (Claude, ChatGPT, Gemini, Copilot)

## Mini-case: one hypothesis, six answers



- Left: 95% intervals for the mean HICP inflation since 2013 (mean 4.30%); right: size of each test in a Monte Carlo with the fitted AR(1),  $\phi = 0.99$ ,  $T = 164$

## Interpreting the mini-case

- A typical assistant proposal: “NW with  $\lfloor 4(T/100)^{2/9} \rfloor$  lags” gives  $t = 2.69$  and rejects  $\mu = 2.5$ 
  - ▶ Its actual size at this persistence: 79% instead of 5%; the naive test: 90%
- NW-Andrews ( $t = 1.17$ ), fixed- $b$  ( $t = 1.59$ , critical value 2.23) and EWC ( $t = 1.63$ ,  $\nu = 12$ ) do not reject
  - ▶ But their sizes are still 58%–63%: a non-rejection is not evidence for the target either
- Honest conclusion: with 14 years of a near-integrated series the data cannot settle the question; that is the open part

## Project idea

- **Question:** which HAR test is reliable for the mean of Romanian and euro-area inflation, 2005–2026?
  - ▶ Replicate first: one size table of [Lazarus et al. \(2018\)](#) (their AR(1) design); explain every gap
- **Extension**
  - ▶ add the tests of [Müller \(2014\)](#); calibrate the Monte Carlo to HICP data of all EU countries (Eurostat)
  - ▶ report size and power on one frontier per country; test the target hypothesis with the test that controls size
- **Deliverables:** a dated pre-analysis plan; a replication package; AI\_USE.md with the prompts that shaped decisions; AI\_ERRORS.md with each invented reference or wrong number caught

## Key takeaways

- ▣ The precision of a mean or a slope is governed by the long-run variance  $\Omega = 2\pi f(0)$ , not by  $\gamma_0$
- ▣ Ergodicity gives consistency; MDS or mixing structure gives the CLT; the HAC estimator gives the standard error
- ▣ Kernel, bandwidth and critical values are one decision: NW with  $1.3\sqrt{T}$  and fixed- $b$ , or EWC, by default
- ▣ Block bootstraps resample dependence; the wild bootstrap does not
- ▣ Near unit roots defeat all of them: measure the size of your test by Monte Carlo at your persistence
- ▣ Searches need search-adjusted  $p$ -values; projects need replication packages and a plan fixed in advance

## Self-assessment

- ▣ What is the long-run variance of an MA(1) with  $\theta = -0.5$  and  $\sigma^2 = 1$ ?
- ▣ Why can the truncated kernel give a negative variance while the Bartlett kernel cannot?
- ▣ Which standard error would you use for the mean of daily S&P 500 returns?
- ▣ Which standard error would you use for the mean of their squares?
- ▣ Why is the wild bootstrap invalid for a regression with overlapping four-quarter growth?
- ▣ You tried 40 specifications and report the best one: which  $p$ -value should you report?

## Next chapter and further reading

- ▣ **Next:** [Chapter 1](#), forecast evaluation, scoring rules and combination
  - ▶ Diebold–Mariano is a HAC  $t$ -test on loss differentials; SPA and the Model Confidence Set use the stationary bootstrap of this chapter
- ▣ **Further reading**
  - ▶ Theory: [Hamilton \(1994\)](#) (Ch. 7 and 10), [Bradley \(2005\)](#), [Lahiri \(2003\)](#)
  - ▶ Practice: [Lazarus et al. \(2018\)](#), [Lazarus, Lewis and Stock \(2021\)](#), [Müller \(2014\)](#)
  - ▶ Snooping: [White \(2000\)](#), [Hansen \(2005\)](#), [Romano and Wolf \(2005\)](#), [Harvey, Liu and Zhu \(2016\)](#)
- ▣ **Code**
  - ▶ Quantlets: [github.com/danpele/Advanced-Time-Series/tree/main/Quantlets/Ch\\_00](https://github.com/danpele/Advanced-Time-Series/tree/main/Quantlets/Ch_00)
  - ▶ Python: `statsmodels (cov_type="HAC")`, `arch.bootstrap (StationaryBootstrap, optimal_block_length, SPA)`

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